

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2025

Bachelor in Computer Applications

Course Title: **Web Technology**

Code No: **CACS205** Semester: **III**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What is client-side and server-side image maps? Explain with example.
3. Differentiate between inline, internal, and external CSS with examples.
4. Explain 2-tier and 3-tier architecture with proper diagrams.
5. What is a Formal Public Identifier (FPI)? Describe its purpose and explain the syntax used to define an FPI with appropriate examples.
6. Define DTD. Why XSLT is it important in real-time XML data processing?
7. What is a web server? Describe the various functions and responsibilities performed by a web server.
8. Explain how sessions are used in server-side scripting to manage user data. Describe how session data differs from cookies and write server-side script to demonstrate session creation, removal and retrieval.

Group C

Attempt any TWO questions[2x10=20]

9. Write HTML code to create a form for registration of customer data for ecommerce website with field (name, email, phone, address, date of birth, gender) and write server-side code to validate form for required validation for all field, type validation for phone and store that information inside "customers" table within "ecommerce" database.
10. Design and create a responsive webpage using HTML and CSS that includes the following features: (Make necessary assumptions for style information): Header section must contain website logo on the left and include a horizontal navigation menu on the right with at least 4 links.i. Content section with two columns layouts first part with 80% width which must have A heading(h1) and a paragraph describing the website or product and image with an alt attribute, second part with 20% width contains list of services company provides.ii. Footer section must contain contact information (address, phone and email) with location map(google) for your company using iframe.iii. Use semantic tags to define different block of page and navigation must be vertical for width below 756px.

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- 11.** Write XML code and create XML Schema Definition (XSD) to validate the structure and data types of the customer XML data. Your schema should enforce the following rules: CustomerID should be a required integer element. Name should be a required string with a maximum length of 100 characters. Phone should be a required string consisting of exactly 10 digits. Address should be an optional string. DateOfBirth should be a required date element with a format YYYY-MM-DD.